

Monica Jin

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Education

North Carolina State University

Computer Science B.S. | Minor in Art & Design | GPA 4.0

Expected Graduation - **May 2027**

Raleigh, NC

Relevant Coursework: Data Structures, Algorithms, Operating Systems, Intro to AI, Computer Graphics

Experience

Salesforce

May 2026 - August 2026

Software Engineer Intern - Event Monitoring & Security

San Francisco, CA

- Designed and shipped audit logs and APIs that surface suspicious GraphQL activity, helping customers detect potential data exfiltration. (**Java, Apex**)
- Added user-agent telemetry across major APIs, enabling customers to identify external AI agents and connected applications; partnered with 8+ product & security teams to deliver data end-to-end.
- Cut customer log-export bandwidth by **90–95%** by building server-side compression, improving reliability for enterprise-scale customer log retrievals.
- Integrated unit and UI visual test suites (**JUnit**) into CI/CD pipelines, using AI-driven REST API testing and synthetic log datasets to reduce regression risk.

IRIS Technologies

August 2025 - May 2026

Software Engineer Intern

Raleigh, NC

- Led software design for movement and object-manipulation behaviors for a humanoid robot, testing in simulation before deploying to hardware. (**Python, C++, NVIDIA Isaac Sim**)
- Built a conversational AI interface for with context-aware responses, demoed live at industry conferences. (**Claude API**)

Pearson

June 2025 - August 2025

Software Engineer Intern - AI/ML Platform

Durham, NC

- Built and deployed a Python image processing pipeline for an LLM platform, integrating GPT-4.1 for image understanding and Azure Content Safety to filter unsafe content, improving document chunk quality by **80%**.
- Introduced video processing as a new platform capability, using MoviePy, Whisper, and GPT-4.1 to generate transcripts and visual summaries.
- Aligned with downstream teams on the ideal output format for image descriptions, then validated against real-world educational samples in CI/CD pipelines.

North Carolina State University

January 2024 - May 2025

Machine Learning Research Intern

Raleigh, NC

- Tested how reliable self-driving car vision systems are when facing tampered images, informing safer perception models.
- Measured a 38.3% accuracy drop across 10,000+ adversarial test images generated via adversarial attacks, exposing robustness gaps in vision transformer models.
- Benchmarked iterative attacks against single-step attacks to compare model robustness under multi-step perturbations, then engineered physical adversarial patches to test attacks under real-world imaging conditions.

Projects

Ceramic AI | Python, Scikit-learn, React, Three.js, TypeScript

[GitHub](#)

- Built a machine learning app that predicts how a glaze will look based on clay and firing conditions, working alongside architecture students who created physical models to validate the predictions.
- Developed a 3D interface that renders predicted glaze finishes on interactive ceramic models in real time.

3D Battlezone | WebGL, JavaScript, GLSL, HTML/CSS

[GitHub](#)

- Recreated Battlezone from scratch with a custom 3D rendering engine, including terrain, combat, and enemy AI.
- Built lighting shaders, mouse-based aiming, and a live minimap using multi-pass rendering.

Technical Skills

Languages: Python, Java, C++, C#, TypeScript, SQL, Kotlin, React, HTML/CSS

AI/ML: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, GPT-4.1, Claude API, Whisper

Developer Tools: Git, Docker, Bazel, Jenkins, CMake, Perforce, Linux, VS Code, IntelliJ, Azure

Graphics & Robotics: Unreal Engine, Unity, Godot, NVIDIA Isaac Lab, Isaac Sim, Figma, Blender